NEW HORIZON COLLEGE OF ENGINEERING
New Horizon Knowledge Park, Ring Road, Marathalli
Autonomous College Permanently Affiliated to VTU, Approved by AICTE & UGC
Accredited by NAAC with 'A' Grade, Accredited by NBA

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

CASESTUDY 1 – GROUP 1
(USN ending with 1, 3,5,7,9)

Course:Computer networks
Sem / Sec: VI A, B, C
Release Date: 07.03.2026

Coursecode: 22ISE62
Max. marks: 15
Due Date :16.03.2026

Q#	Question	RBT Level	Cos	POs & PSOs	Marks
1.	**Scenario 1: Rural Health Monitoring Network** A public health organization plans to deploy a **remote health monitoring system** in rural areas where internet connectivity is limited. Wearable sensors worn by patients collect health data such as heart rate, temperature, and blood pressure. These devices must transmit data to	L3,L4	CO1	PO1, PO2, PO3, PO4, PO5, PO12, PSO1, PSO2	7.5

	nearby mobile health workers' devices using a **peer-to-peer communication network**. The system should continue functioning even if connectivity to the central hospital server is temporarily unavailable. **Case Study Questions** 1. What type of **peer-to-peer architecture** would best support data exchange between wearable devices and mobile health workers in areas with limited connectivity? Explain your choice. 2. How should the system **manage data synchronization and buffering** when the connection to the central server becomes available again?				

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	3. What **routing strategy** would ensure reliable transmission of patient data between multiple mobile nodes in rural areas? 4. How can the system ensure **data privacy and secure communication** of sensitive health information between devices? 5. What **network challenges** such as latency, bandwidth limitations, or node mobility must be addressed in this healthcare monitoring network?				
2.	**Scenario 2: Smart Transportation Vehicle Communication**	**L3,L4**	**CO1**	**PO1, PO2, PO3, PO4, PO5, PO12, PSO1, PSO2**	7.5

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	A city government is implementing a **vehicle-to-vehicle (V2V) communication system** to improve traffic management and road safety. Vehicles equipped with communication modules exchange real-time data such as traffic congestion alerts, accident warnings, and road condition updates. The system must operate efficiently even without continuous connectivity to a central traffic management server. **Case Study Questions** 1. Which **network communication model** would best support direct data exchange between moving vehicles and why? 2. How can vehicles **discover nearby nodes and maintain**				

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	communication while moving at high speeds? 3. What **routing and data dissemination techniques** should be used to quickly spread emergency alerts among nearby vehicles? 4. How can the system ensure **authentication of messages** to prevent malicious or false traffic information? 5. What **trade-offs between latency, scalability, and network overhead** must be considered in such a vehicular communication system?				

Note: Minimum 15 pages report must be submitted.

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CASESTUDY 1 – GROUP 2
(USN ending with 0,2,4,6,8)

Course: Computer networks
Sem / Sec: VI A, B, C
Release Date: 07.03.2026

Coursecode: 22ISE62
Max. marks: 15
Due Date: 16.03.2026

Q#	Question	RBT Level	COs	POs & PSOs	Marks
1.	**Scenario 1: Campus Collaborative File Sharing System** A university wants to develop a **decentralized file-sharing platform** that allows students to share large project files and research datasets directly with each other without relying heavily on the university servers. The system will use **peer-to-peer networking** to distribute files	L3,L4	CO1	PO1, PO2, PO3, PO4, PO5, PO12, PSO1, PSO2	7.5

	efficiently among multiple devices across the campus network. **Case Study Questions** 1. Which **peer-to-peer network model (centralized, decentralized, or hybrid)** would be most suitable for this campus file-sharing system? Justify your answer. 2. How should the system **locate and retrieve files from multiple peers** in the network efficiently? 3. What **methods can be used to improve download performance** when many users request the same file simultaneously? 4. How can the system ensure **file integrity and prevent distribution of corrupted or malicious files**?				

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CASESTUDY 1 – GROUP 2

	5. What **scalability and bandwidth management challenges** might arise as the number of users grows?				
2.	**Scenario:** A startup named **LiveLink** is building a real-time video conferencing application aimed at users in rural and remote areas with limited internet bandwidth. The founders want to avoid reliance on centralized servers to minimize infrastructure costs and improve performance. They plan to use a peer-to-peer architecture with WebRTC to enable real-time video and audio communication between users. However, they are concerned about issues like NAT traversal, data security, scalability, and reliability. **Case Study Question:** *As a network systems consultant, how would you design a scalable and secure real-time peer-to-peer architecture for LiveLink's application? Consider the*	**L3,L4**	**CO1**	**PO1, PO2, PO3, PO4, PO5, PO12, PSO1, PSO2**	**7.5**

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CASESTUDY 1 – GROUP 2

	following in your response: 1. What components and technologies would you recommend (e.g., WebRTC, STUN/TURN servers)? 2. How would you address NAT traversal and firewall issues in peer-to-peer connections? 3. What measures would you implement to ensure end-to-end security and user privacy? 4. How would you balance P2P communication with the need for occasional centralized fallback (e.g., for group calls or connectivity failure)? 5. What limitations might arise from relying purely on a P2P model, and how can they be mitigated?				

Note: Minimum 15 pages report with soft binding must be submitted .